

CHAPTER SIX

MEDICAL SONOGRAPHY

6.1- Ultrasound waves:

They are sound waves of frequencies above the audible sound with very high frequency that can penetrate deeply in the body where audible sound waves cannot do. (And those of frequencies below the audible sound are called infrasound).

The ultrasound is called sonar, an acronym of **s**ound **n**avigation and **r**anging.

The first practical application of sonar was the unsuccessful search for the Titanic ship which had sunk in 1912. But the first important successful application of sonar was in the World War II, and the first success of using sonar in medical diagnosis was done by the Austrian neurologist, Karl Theodore Dussik, (1908-1968), in the late of 1930's.

In medical ultrasound range, the velocity of transmission depends primarily on the physical makeup of the tissue through which the ultrasound is being transmitted and does not depend on the frequency of the wave. The velocity of ultrasound in the soft tissues ≈ 1540 m/s (like that in water), and it depends on the characteristics of the medium.

6.2- Medical sonography technique:

Medical sonography, also known as sonography or ultrasonography, is the uses of high frequency ultrasound waves for medical diagnosis, and it is a branch of the medical diagnostic imaging techniques. The ultrasound waves make echoes which form images of the intended object on a computer screen. (Sec 6.6). The basic idea of sonography is to get ultrasound image, called sonogram or ultrasonogram.

The sonogram is highly dependent on the velocity and intensity of the ultrasound waves.

1- Sonogram performance:

The sonogram is defined as a computer picture of the intended object inside the body by high energy ultrasound waves, which are bounced off the internal organs or tissues, and make echoes.

Sonograms can be performed as follows:

I- Generation of high frequency ultrasound pulses (waves) by piezoelectric or ferroelectric crystals (transducers), where the, amplitude and duration of the pulses can be changed by a pulse control.

II- Transmission of these signals in the skin and tissues of the body to the intended object.

III- Reflection of the ultrasound waves from the intended object (reflected echoes).

IV- Receiving the reflected echoes by the transducers, and generation of electrical signals relative to these echoes.

V- Processing these signals by an advanced computer, and displaying the images on a screen, which can be recorded on a CD and papers. (Fig 6.1).

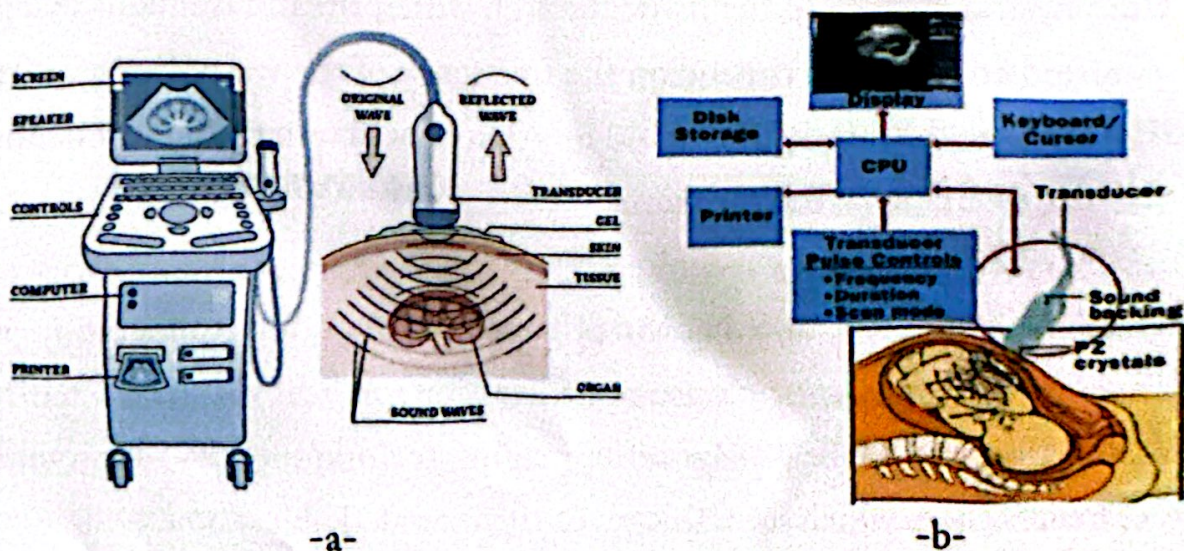


Fig 6.1: Sonography. a- equipment of ultrasound imaging. b- schematic diagram of a sonography.

2-Tissue characteristics affect ultrasound velocity:

The velocity of ultrasound in the tissues plays a big role in the medical sonography.

The most important characteristics of the transmitting tissue material which affect the velocity of ultrasound waves are the compressibility and density of the transmitting tissue material.

I- Compressibility of the tissue material:

Compressibility is the measure of rate of change of a material volume in response to the rate of pressure change. When the tissue material is less compressible, it more rapidly transmits ultrasound. Hence, the ultrasound velocity in the body is faster in the bones (≈ 4080 m/s) because their molecules are closer to each other, less fast in the blood (≈ 1540 m/s), water, and other liquids, because their molecules are closed to each other but less than that in solids, and slowly in gases (≈ 330 m/s), such as the lungs, because the gas molecules are far apart and easily compressed. Therefore, the velocity of ultrasound (v) is inversely proportional to the compressibility of the conducting tissue material.

II-Density of the tissue material:

The higher density material of large atomic mass molecules cannot transmit ultrasound as great as materials of the same state composed from smaller atomic mass molecules. for example, in liquid state, the transmission of sound in water (atomic mass = 18) > that in mercury (atomic mass = 200.59).

In any medium, the velocity of ultrasound (v) is inversely proportional with the density ρ and the compressibility k of the medium, and it can be given by:

$$v = \sqrt{1/\rho k} \quad \dots(6.1)$$

The frequency of ultrasound is constant in any particular medium, and when the frequency increases, the wavelength must decrease.

3- Ultrasound beam intensity:

The intensity of ultrasound beam (or loudness in the audible range) is defined as the sound power per unit area normal to the direction of propagation. Also, it can be defined as the product of the wave pressure (p) by the wave velocity, i.e.:

$$I = p v \quad \dots (6.2)$$

The ultrasound beam intensity can be described by the relative intensity level and beam attenuation.

I-Relative intensity level:

The relative intensity level β of an ultrasound beam, the same that of sound, is the level of ultrasound intensity (I) relative to a reference intensity (I_0), measured in Bell (B) or deciBell (dB) and it is not the same as the actual intensity, where:

$$\beta (B) = \log \frac{I}{I_0} \quad \text{or,} \quad \beta (dB) = 10 \log \frac{I}{I_0} \quad \dots (6.3)$$

Example 6.1:

An ultrasonic beam of original intensity = 100 W/cm^2 incident on the liver for liver scanning, and the intensity of reflected echo from the liver = 0.01 W/cm^2 . Calculate its relative intensity level.

Solution:

The relative intensity level can be calculated using eq 6.3:

$I_0 = 100 \text{ W/cm}^2$ and $I = 0.01 \text{ W/cm}^2$ Thus:

$$\beta (B) = \log 0.01/100 = -4, \text{ and}$$

$$\beta (dB) = 10 \log 0.01/100 = -40 .$$

The Bell or decibel can be positive or negative, where the positive bell or decibel means a gain of energy and the negative means a loss of energy.

Ultrasound loses energy when it passes through tissues, so in the above example, the intensity of the returning echo relative to the original beam is 4B or 40 dB.

II- Beam attenuation:

It is the reduction in the beam intensity by reflection (or scattering), refraction, and absorption of the waves from the ultrasound beam, when an ultrasound beam incident on a boundary between two organs or tissues in the body. The reflection of the waves occurs by a homogenous medium and the scattering occurs by a heterogeneous medium.

The beam intensity decreases exponentially with the thickness (x) of the tissue, and it can be given by:

$$I = I_0 e^{-\mu x} \quad \dots (6.4)$$

Where μ is the linear attenuation coefficient of ultrasound beam by the tissue, and the unit of μ is m^{-1}

4- Medical applications of sonography:

There are several medical applications of sonography in diagnosis of the body such as:

I- Creation images of specific areas in the body, diagnosis of various diseases, provide diagnosis information, and records anatomical conditions.

II- Diagnosis of some internal body structures like the internal organs, muscles, joints, and identifying the size and structural changes in the pelvic organs.

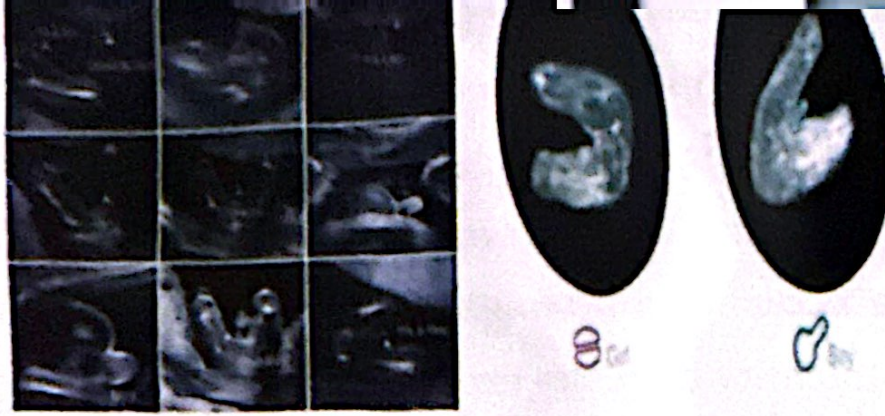
III- Diagnosis of some obstetrical and gynecological cases like:

a- Determination of the pregnancy, where it can identify a healthy pregnancy at 17 days after the egg was released from the ovary (ovulation).

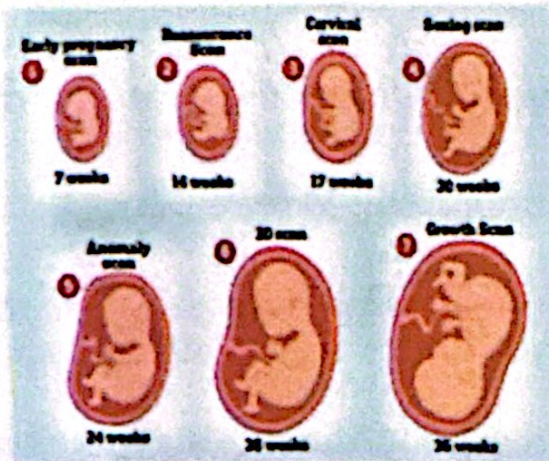
b- Determination of the fetus sex (at weeks 11- 20), where a caudal notch in female and a cranial notch in male are demonstrated in examination of the genital region in the midline sagittal plane. (Fig 6.2a).

c- Monitoring the health and development of the fetus where the pregnant women can have 7 scans during pregnancy. (Fig 6.2b).

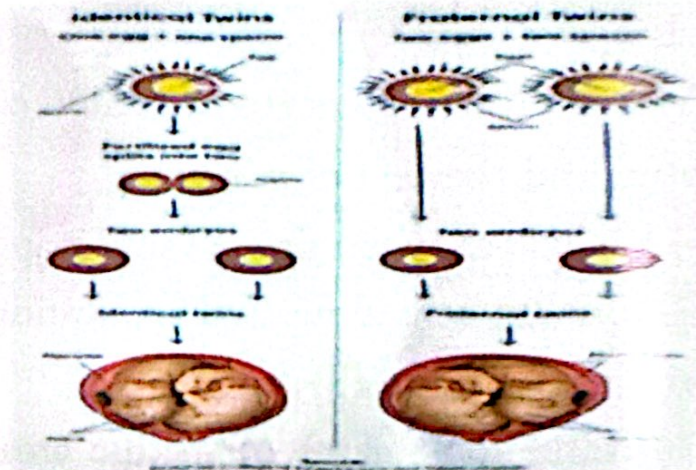
d- Diagnosis of twin (or multiples) pregnancies where the twins (or multiples) are either identical formed from single egg-single sperm, or fraternal (may not have the same sex or appearance) formed from dual (or multiples) eggs - dual (or multiple) sperms. (Fig 6.2c).



-a-



-b-



-c-

Fig 6.2: Fetal sonography. a- determination of fetal sex. b- pregnancy required scans (monitoring of fetal development). c- twin (or multiples) pregnancy scan.

e-Evaluation of the fetal size, maturity, placental positions and help to pick up any fetal abnormality.

Sonography is a reliable fetal test even though it does not give 100% accuracy, and when fetal abnormalities are detected, further tests are required to confirm the diagnosis such as amniocentesis and chronic villus sampling.

(Amniocentesis is a test that takes amniotic fluid from around the fetus in the womb, and chronic villus sampling is a prenatal test obtained either by passing a small catheter through the vagina and cervix in the womb, or by inserting a fine needle through the wall of abdomen).

IV- Examination of the heart and blood vessels.

V- Identifying and distinguishing cancers and benign cysts, and evaluation of the tumors.

VI- Evaluation of foreign bodies of the eye, and demonstration of retinal detachment.

5- Advantages of sonography:

There are several advantages of using sonography on the other prominent methods of medical imaging that the sonography:

I- Does not use harmful ionizing radiation.

II- Lower in cost.

III- Portable and can be brought to the bedside.

IV- Allows making a faster decision.

V- Used in special diagnosis, e.g., they are used experimentally in biomicroscopic techniques in special regions such as anterior chamber of the eye, where the range of the used frequencies = 50-100 MHz.

6.3- Ultrasound generation crystals:

Ultrasound waves are generated by certain kinds of very thin crystals elements which have special designs. The crystal oscillates with the required frequency of sonography and gives only one frequency.

The crystals, the most important component of the ultrasonic scanner, are small energy resources of very small thicknesses of few millimeters, located near the face of the probe and encased in a plastic housing. (Sec 6.8.4).

The crystal is a transducer that can convert one form of energy into another. Ultrasonic crystals are employed to convert a.c electric signals (electric energy) into high frequency ultrasonic waves (mechanical energy) that can be transmitted into the tissues, and convert the reflected ultrasonic energy (echo) from the higher density tissues into a.c electric signals in the crystal.

1- Piezoelectric and ferroelectric crystals:

The sonography crystals mainly, are either piezoelectric (piezo is a Greek word means pressure) or ferroelectric crystals.

Piezoelectric crystals are naturally occurring materials possess piezoelectric properties, e.g., quartz. But most of the crystals used in medical ultrasound are manmade ceramic ferroelectric crystals which have replaced the piezoelectric

crystal materials that were used earlier. The synthetic ceramic crystals may be soon replaced by plastic polymers in the construction of ultrasound transducers.

The main difference between the piezoelectric and ferroelectric crystals is that the polarization of piezoelectric crystal can be induced by application of electric field, while in the ferroelectric the polarization is induced spontaneously without application of electric field, as the center of +ve charge naturally does not coincide with the center of -ve charge.

In older versions of ultrasonic transducers, either one crystal is used as a transmitter and a receiver, but it cannot transmit and receive the ultrasonic waves at the same time, or one crystal is used as a transmitter and another crystal as a receiver. But recently multi crystal elements are used as transmitters and receivers.

2- Ultrasound crystal design:

The crystal is designed to generate high frequency ultrasound waves applying an electric field, where its thickness (x) exactly equals to half of the wavelength (λ) of the ultrasound wave to be produced by this crystal, i.e., $\lambda = 2x$.

A large variety of sizes and shapes of crystals are available to perform specific functions of the sonography, but they all have a general design. The shape of crystal is a hexagonal, and it changes its size and shape when either a strong short electrical pulse from a.c voltage source, or the echo of ultrasound waves, is applied on the crystal.

The front and back faces of the crystal are coated with a thin, very good conducting film of gold or silver electrodes, to ensure very good contact with the two electrodes that supply the a.c voltage. An outside electrode is grounded, and it is connected to an outside surface of the crystal which is coated with a very good electrical insulator to protect the patient from electrical shock.

3- Crystal oscillation:

The applied voltage drives the crystal to oscillate at a certain frequency, depending on the crystal thickness. The thinner is the crystal, the more frequency ultrasonic waves are generated, and the more price is the crystal.

The crystal oscillates more efficiently (i.e., resonates) at the frequency determined by its thickness.

The frequency of the crystal that corresponds to half of the wavelength of the produced ultrasonic beam is called fundamental resonance frequency of the crystal.

Example 6.2:

Calculate the fundamental resonance frequency of a lead zirconate titanate (PZT-4) crystal of 1 mm thickness, where the speed of sound in PZT-4 = 4000 m/s.

Solution:

The crystal will resonate at the fundamental frequency when the wavelength equals twice of the crystal thickness i.e.:

$$\lambda = 2x$$

$$\lambda = 2 \times 1 \text{ mm} = 2 \text{ mm} = 0.002 \text{ m}$$

In general, the frequency f of a wave is given by:

$$f = v / \lambda \text{ thus:}$$

$$\text{Fundamental frequency} = \frac{v}{\lambda} = \frac{4000 \text{ m/s}}{0.002 \text{ m}} = 2 \text{ MHz}.$$

Hence, a 2 MHz piezoelectric crystal made of PZT-4 will have a thickness = 1 mm, and similarly, 1 MHz piezoelectric crystal will be of a thickness = 2 mm.

6.4- Parameters affect the transmission of ultrasound waves in the skin and tissues:

When an ultrasound beam incident on the skin of the body, the skin reflects > 99.9% of the ultrasound beam generated from the transducer, and < 0.1% of the beam is transmitted. Therefore, the reflected echo from the intended object is very small and cannot make any image of any object in the body. The high reflection of the skin happens because the ultrasound waves emitted from the transducer are transmitted through the air to the skin, where the acoustic impedance of the air << than that of the skin.

The transmission of ultrasound beam through the skin and the other tissues inside the body can be highly affected relative to three important factors, the acoustic impedance, lubricating gel, and angle of the beam incidence.

1- Acoustic impedance:

Acoustic impedance, a fundamental property of the matter, is the effective resistance that an ultrasound beam encounters as it passes through the tissue. i.e., the opposition to the transmission of sound waves.

The acoustic impedance (Z) depends on the tissue density and speed of sound waves in the tissue, and can be given by:

$$Z = \rho v \quad \dots (6.5).$$

The unit of impedance is $\text{gm/cm}^2.\text{s}$ (in cgs system), or $\text{kg/m}^2.\text{s}$ (in MKS system), where $1 \text{ gm/cm}^2.\text{s} = 10 \text{ kg/m}^2.\text{s}$.

It is found that it will be more convenient to change the unit of impedance to Rayl, where $1 \text{ Rayl} = 1 \text{ kg/m}^2.\text{s}$.

Rayl is named in the honor of the British physicist Lord John William Strutt, 3rd Baron Rayleigh, (1842-1919).

Example 6.3:

Calculate the acoustic impedance of a soft tissue where the velocity of sound in the soft tissue, $= 1540 \text{ m/s}$, and the density of tissue $= 1.05 \text{ gm/cm}^3$.

Solution:

$$\text{Density of soft tissue} = 1.05 \text{ gm/cm}^3 = 1050 \text{ kg/m}^3$$

As; $Z = \rho v$, therefore:

$$1050 \times 1540 \text{ kg/m}^2.\text{s} = 1.62 \times 10^6 \text{ kg/m}^2.\text{s} = 1.62 \text{ MRayl}.$$

As the velocity of sound in tissue is fairly constant over a wide range of frequencies, therefore the acoustic impedance is constant for these tissues.

When ultrasound waves pass from the air to skin, or from one tissue plane to another inside the body, the amount of reflected beam is determined by the difference in the acoustic impedances of the two media or tissues. The reflection happens only when the impedance of the reflecting tissue is different from that through which the beam transmitted. The greater the difference between the acoustic impedances of the two media or tissues, the greater is the percentage of reflected beam.

The differences of acoustic impedances of most body tissues are small except that of air (relatively very small), and bone (relatively very large). (Table 6.1).

Table 6.1: Acoustic impedance of some body materials, tissues, and organs.

Type	Density(kg/m ³)	Sound velocity (m/s)	Acoustic impedance (MRayl)
Air *	1.225	337.23	0.0004
Water	997	1480	1.4800
Soft tissue	1050	1540	1.6200
Fat	900	1440.2	1.3000
Lung	240	47.5	1.1400
Brain	1046	1546.3	1.6200
Blood	1050	1500	1.5400
Muscle	1100	1588.4	1.7500
Liver	1079	1585.7	1.7100
Skin	1109	1624	1.8000
Stomach	1088	1500	1.6300
Kidney	1066	1554.3	1.6600
Cartilage	1100	1639.6	1.8000
Spleen	1089	1567.6	1.7100
Bone:			
(cancellous)	1178	2117.5	2.4900
(compact)	1908	3514.9	6.7100
Tooth	2180	4565.9	9.9500

* The air is at STP (standard temperature and pressure).

** The range of sound speed in the adult healthy lungs = 25-70 m/s (numerical average = 47.5 m/s).

2- Lubricating gel:

The very high reflection of sonar beam from the skin requires a specific lubricating gel of an acoustic impedance equals to that of the skin to lubricate the skin of the patient. It makes a coupling between the transducer phase and the skin, eliminates the air layer, and highly reduces the reflected beam from the skin.

The sonar lubricating gel should have an acoustic impedance equals to that of the soft tissue i.e., there should be no difference between the acoustic impedances of the gel and skin, in addition to other important characteristics of the gel.

The use of ultrasound gel between the surfaces of the transducer and skin of the patient is an effective way to reduce the beam reflection, increase the beam transition, increase the echoes from the tissue back to the transducer, and produce a good image.

3- Angle of beam incidence:

The angle of incidence between the ultrasound beam and reflecting surface of the intended object plays very important role in the amount of transmitted beam in the skin.

When an ultrasonic beam incident on the intended object through the skin and body tissues, and the angle of incidence is large, therefore, less beam is reflected by the intended object and more beam is transmitted through it, which gives a worse sonogram.

The best reflection occurs when the angle of incidence = 90° , i.e., there is a total internal reflection and no transmitted waves through the intended object. But no adequate reflection of ultrasound happens when the angle of incidence $> 3^\circ$ from the perpendicular. Therefore, the expert of the sonographic examiner is very important to obtain a good image by sonography.

The refracted beam in the intended object, depends on the angle of incidence of the beam on the intended object, and the difference in acoustic impedances between the two media, the intended object and tissues surrounding it.

When ultrasound beam transmits from one medium to another, it will result in a change in the direction of refractive waves. This change necessitates a change in the wavelength of these waves to accommodate a new velocity in the second medium, but its frequency remains constant.

When the velocity of ultrasound wave is lower in the second medium (more acoustic impedance medium), therefore, the angle of refraction $<$ angle of incidence.

The transmitted beam (T) in the medium is given by:

$$T = \frac{4Z_1Z_2}{(Z_1+Z_2)^2} \times 100\% \quad \dots (6.6)$$

Where Z_1 = acoustic impedance of the medium 1 (tissues), and Z_2 = acoustic impedance of the medium 2 (intended object).

6.5- Echo (ultrasound reflection):

Echo is the reflected ultrasound waves from the intended object back to the crystal (or crystals) which converts these waves to electric signals, and forms the sonogram.

The beam can be reflected directly back to the transducer in an angle depends on the angle of incidence on the intended object, where the angles of incidence and reflection are equal. Echo, which is received back by the crystal, measures the time interval between sending the signal and receiving the echo to determine the distance of the intended object.

The amount of echo at nearly perpendicular incidence on the surface of the intended object is given by:

$$R = \left(\frac{Z_2 - Z_1}{Z_2 + Z_1} \right)^2 \times 100\% \quad \dots (6.7)$$

Where R= percentage of the reflected beam, and the rest of the beam is transmitted through the intended object.

The incident beam is not totally reflected from the intended object, because the angle of incidence $\neq 90^\circ$ through all the time of scanning. Thus, a certain percentage of the beam is reflected, and the other percentage is transmitted (refracted) through the intended object, and:

$$R + T = 100\%$$

This means that, in the intended object, when the transmitted beam is increased, the reflected beam is decreased, and vice versa.

Example 6,4:

Calculate the percentage of reflected and transmitted ultrasound beams in going from the lung of a patient ($Z = 0.0004$ MRayl) to his chest wall ($Z = 1.7$ MRayl).

Solution:

$$R = \left(\frac{Z_2 - Z_1}{Z_2 + Z_1} \right)^2 \times 100\% = \left(\frac{1.7 - 0.0004}{1.7 + 0.0004} \right)^2 \times 100\% = 99.9\%$$

$R + T = 100\%$. Thus:

$$T = 100\% - R = 100\% - 99.9\% = 0.1\%$$

Also, T can be calculated using eq 6.6:

$$T = \frac{4Z_1Z_2}{(Z_1+Z_2)^2} \times 100\% = \frac{4 \times 1.7 \times 0.0004}{(1.7+0.0004)^2} \times 100\% = \frac{0.00272}{2.89136016}$$

$$T = 0.0009 \cong 0.001 = 0.1\%$$

$$\text{Therefore, } R+T = 99.9\% + 0.1\% = 100\%$$

6.6- Sonography image display:

The sonogram is dependent totally on the echo reflected from the intended object and incidents on the crystal elements, where more echo incidence means clearer displayed image.

1- Echo reception by the transducer:

This process can be summarized as follows:

I- Echoes are reflected from each tissue interface when the ultrasonic pulses pass through the body, back toward the transducer, and incident on the crystals.

II- The energy of reflected echo waves which incident on the crystal causes a mechanical compression on the crystal element and induces a voltage between the crystal electrodes (mechanical energy changes to electrical energy).

III- The induced voltage is amplified and displayed as electric signal on the screen.

2- Image display:

The image display can be achieved by:

I- Display of the ultrasound images of an object in the body on a screen by pixels (the smallest controllable element of a picture represented on the screen), relative to the height and intensity of echo signals. The image display takes a time called image time.

II- The image time is the time that the beam takes to transmit from the transducer into the target, reflected from the target, and detected by the transducer as echo waves from the target.

III- Processing of the produced electric signals by a sophisticated computer to form an image which is displayed on the screen of ultrasound scanner.

6.7- Factors affect the absorption of ultrasonic beam:

Absorption of ultrasound beam is the removing of ultrasound waves i.e., ultrasound energy, from the beam due to the acoustic impedance and the frictional forces that oppose the motion of particles in the tissues. The removed energy is converted to heat, and thus the absorption refers to the conversion of ultrasonic energy (mechanical energy) to a thermal energy.

The absorption coefficient, usually employed in absorption of ultrasound beam, can be defined as the reduction of relative intensity level of ultrasonic beam per unit distance. The unit of absorption coefficient is (dB/cm).

The amount of absorption of ultrasound beam in a medium is determined by viscosity of the conducting medium, relaxation time of molecules of the conducting medium, and frequency of the ultrasound beam.

1- Viscosity of the conducting medium:

The importance of viscosity of the conducting medium is obvious in ultrasound beam absorption because the ultrasound beam is transmitted by the vibrating particles of the medium. The absorption of ultrasound beam is dependent on the medium viscosity, whether the viscosity is high, medium, or low.

I- High viscosity:

When the viscosity increases, like that in the bone, the motion freedom of the particle decreases and the internal friction increases. This will lead to more absorption of energy from the beam, by reduction of intensity of the ultrasound beam, and converting the absorbed ultrasound energy into heat.

II- Medium viscosity:

In medium viscosity media like soft tissues, blood, water, and other liquids of the body, the absorption of ultrasound beams is medium, because the motion of molecules is not very restricted.

III- Low viscosity:

In low viscosity media like the lungs and gases in the body, the viscosity is very low, and the absorption is extremely low, because the movement of air molecules are nearly free.

2- Relaxation time:

The relaxation time is the time that a displaced molecule takes to return back to its original position after it has been displaced, and it is constant for any particular medium. It is possible that two media with the same viscosity may have different relaxation time.

When a molecule of a short relaxation time is pushed by an ultrasound wave, it has a time to return back to its resting state before the next compression wave arrives, and no energy will be lost. While the molecule of a long relaxation time may not be able to return completely before a second compression wave arrives. In this case, the molecule moves in the opposite direction of the wave, and more energy from the ultrasound wave is required to reverse the molecule direction.

3- Frequency of ultrasound:

In the soft tissues, there is a linear relationship between the absorption of ultrasound waves and frequency. The absorption is nearly doubled when the frequency is doubled, and it is halved when the frequency is halved.

Selection of the correct transducer for a particular diagnostic job is dependent on the knowledge of ultrasound frequency, i.e., the waves absorption.

High frequency beams have an advantage over low frequency beams, where the depth resolution is superior with the high frequency. But high frequency beams have a disadvantage, that the absorption of ultrasound waves increases when the frequency increases, because higher frequency ultrasound have shorter waves that are easily absorbed or attenuated by the tissue.

Typical sonographic instruments operate in frequencies $\approx 1 - 20$ MHz. Also, frequencies of ultrasound up to 100 MHz can be used in special diagnosis.

Relatively low frequency beam is required to penetrate thick parts of the body, but with less resolution, and can image the deeper parts in the body.

For this reason, the lower frequencies are used for abdominal diagnosis, and higher frequencies for superficial diagnosis. (Table 6.2)

Table 6.2: Guide to typical frequencies used in sonography.

Ultrasound frequency (MHz)	Types of imaging
2.5	1- Deep abdomen. 2- obstetric and gynecological
3.5	1- General abdomen. 2- obstetric and gynecological
5	1-Vascular. 2- breast. 3- pelvic
7.5	1- Breast. 2- thyroid
10	1- Breast. 2- thyroid. 3- superficial veins. 4-superficial masses. 5- musculoskeletal
15-20	1-Superficial structures. 2- musculoskeletal

6.8- Construction of the probes of ultrasound scanner:

The probe of ultrasound scanner, also called transducer as it contains the crystals, is the handheld part of the ultrasound device of a length = 5-15 cm. The frequency and angle are adjustable to send the ultrasonic wave

Nowadays, the most commonly sonographic techniques used in medicine, are the Doppler (pulsed and continuous waves) and echocardiography.

6.12- Doppler sonography:

Doppler sonography is very important medical technique based on the Doppler effect phenomena, and employed to perform very accurate imaging

of the movement of blood, fluids, and tissues, and their relative velocities to the ultrasound probe.

1- Doppler effect:

Doppler effect (or Doppler shift) is a change in the perceived frequency of ultrasound waves reflected by the moving source.

The difference between the two frequencies (frequencies of incident and reflected waves), usually falls in the audible range for most of the physiological motions. This frequency difference is amplified and becomes audible signals.

When the ultrasound source is moving toward the probe, the frequency of ultrasound increases and the wavelength decreases, and vice versa. The ultrasound velocity does not be affected by the source motion, so the ultrasound moves at the same velocity in all directions.

When an ultrasound beam of a velocity v incident on the blood in an artery of a flow velocity u , away from the ultrasound beam, with an angle of incident θ , the blood sends back scattered echoes with a shift in the frequency of the received echo called Doppler shift (f_d), where:

f_d = reflected frequency- transmitted frequency.

Therefore, Doppler shift can be given by:

Doppler shift = $(2 \times \text{velocity of blood} \times \text{original frequency of the beam} \times \cos \theta) / (\text{propagation speed})$.

Thus:

$$f_d = 2u f_0 \cos \theta / v \quad \dots (6.9)$$

Where f_0 = original frequency of the beam.(Fig 6.7).

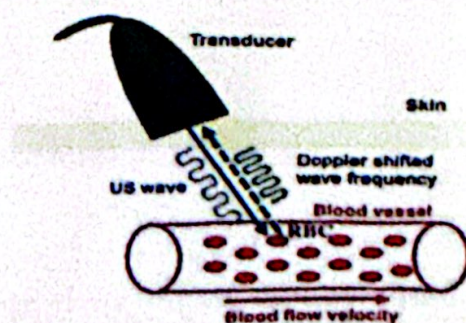


Fig 6.7: Doppler sonography of the blood.

The integer 2 in the above equation means that there are 2 Doppler shifts during the clinical sonographic test.

In Doppler sonography, the ultrasound frequency is changed and the speed stays constant, opposite of that in the regular sonography.

In many cases of sonographic medical scanning, the phase shift of the wave is measured instead of Doppler shift of the received signals, when the received signals arrive.

2- Medical applications of Doppler sonography:

Doppler technique is used widely in three major fields of medicine, the blood flow in the arteries and veins, obstetrics and gynecology, and echocardiography.

Doppler sonography can produce several accurate medical assessments such as:

I- Direction and velocity of blood flow in the arteries and veins, which can be used as an effective tool for diagnosis of vascular problems like stenosis. In this measurement there is only one limitation, that is the ultrasound beam parallel to the blood flow as much as possible.

II- Fetal heart rate.

III- Entry point of umbilical cord into the placenta.

IV- Bleeding due to misplaced placenta (placenta previa), where the fetus placenta covers the cervix of his mother partially or totally.

V- Intrauterine (within the uterus) transfusion for the Rhuses (Rh) incompatibility. Rh incompatibility is a condition that occurs when a pregnant woman has Rh-negative blood, and the baby in her womb has Rh-positive blood, where Rh-positive refers to the blood having Rh factor, which is a protein on the red blood cells.

VI- Detection of other places of uterus.

3- Technical tests of medical Doppler sonography:

There are several technical tests of Doppler sonography such as color, power, spectral, and duplex Doppler.

I- Color Doppler:

It is a Doppler sonography utilizes a sophisticated computer to change the ultrasound waves into certain colors to show the speed and direction of the blood flow in real time. (Fig 6.8 a).

II- Power Doppler:

It is a color Doppler in which more details of the blood flow can be given but it cannot provide the blood flow direction. (Fig 6.8b).

III- Spectral Doppler:

In which the blood flow can be shown on a graph to investigate how much blockage exists in a blood vessel. (Fig 6.8c).

IV- Duplex Doppler:

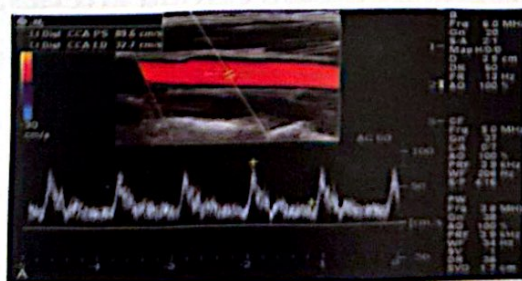
In which the images of blood vessels and organs can be given, then the computer plots these images as a graph. (Fig 6.8d).



-a-



-b-



-c-



-d-

Fig 6.8: Doppler sonography. a- color Doppler. b- power Doppler. c-spectral Doppler. d- duplex Doppler.

6.13- Continuous and pulsed wave Doppler sonography:

There are two modes of Doppler techniques which are the continuous and pulsed wave Doppler modes.

1- Continuous wave Doppler (CW Doppler):

It employs two sets of multi crystal elements in a single head, one is a transmitter and the other is a receiver. In the old versions of this scanner, two crystals are used instead of the two sets of crystal elements.

The transmitter crystal elements (or crystal) transmit continuous ultrasonic signals at a certain frequency, and the receiver crystal elements (or crystal) receive the returning echoes and record their frequency. (Fig 6.9a).

Doppler information is sampled along a line through the body, and all velocities detected at each time point are presented on a timeline.

CW Doppler can measure very high velocities, but it cannot pinpoint where they come from on the scanline, i.e., it does not determine where the ultrasonic wave is reflected from.

In this mode, the Doppler shift is greatest when the beam strikes a blood vessel at a small angle, where the smaller the angle the greater is the Doppler shift. The Doppler shift is amplified and can be heard by the human ear.

CW Doppler is extremely sensitive to weak signals; therefore, it can employ high frequency ultrasound up to 10 MHz. Thus, it can detect the superficial objects such as the arteries within the eye, female breast, carotid arteries, and vessels in the limbs, but it cannot be used in obstetrics, abdominal, and deep objects examination.

Low frequency CW Doppler is usually used in the monitoring of the fetal heart sounds in the womb of the pregnant woman.

CW Doppler instruments provide velocity information but lack of any form of depth resolution, thus it is not useful in the heart imaging.

2- Pulsed wave Doppler (PW Doppler):

It employs a transducer of a single set of multi crystal element used as a

transmitter and a receiver. The ultrasound waves in this mode are short bursts emitted repetitively at a precisely controlled rate into the tissues. A new pulse is not transmitted until the echo from the previous pulse have been detected.

PW Doppler information is sampled from only a small volume defined in 2D image and presented on a timeline. The burst period $\approx 0.5 - 1 \mu s$ which can provide a good axial resolution with interfaces separation $\approx 1 \text{ mm}$.

PW Doppler provides detection of slower velocities at a specific depth where the returning signal is originated, and information of both velocity and position simultaneously.

The depth at which a returning signal is originated can be determined by the knowledge of the time of signal transmission and the time of its return.

In PW Doppler, the operator selects Doppler signals from any depth along the transducer axes. He controls the depth from which the Doppler shift signal will be received, and the axial length along the beam from which the signal will be read. (Fig 6.9b).

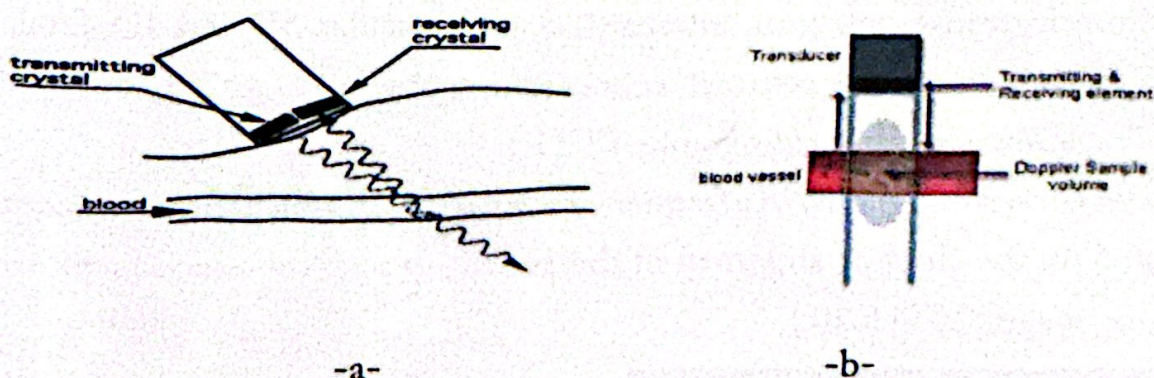


Fig 6.9: Signals of Doppler sonography. a- continuous. b- pulsed.

PW Doppler allows evaluation of the nature of blood flow in normal and stenotic vessels, and evaluation of the heart and heart valves problems in a technique called echocardiography.

1. Echocardiography (echocardiography):